Daniil

DATA SCIENTIST / MACHINE LEARNING ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English — B2

Domains
Management
Automations
Analytics

Data Scientist / ML Engineer with more than 3 years of experience.
Data Scientist with a sharp focus on data-centric projects. Proven ability to identify business problems and address them using Data Science techniques. Proficiency with end-to-end ML projects lifecycle: from data collection to model deployment. Areas of expertise: Natural Language Processing and Generative AI.

Programming Languages
Python.

Data Science
Numpy, Pandas, Matplotlib, Seaborn.

Machine Learning
Scikit-learn, XGBoost, LightGBM, Prophet, Boruta.

Deep Learning
PyTorch, Hugging Face, BitsAndBytes.

Computer Vision
OpenCV, Tesseract OCR, Ultralytics.

Natural Language Processing
NLTK, OpenAI API, Spacy, Langchain, LlamaIndex, CoreNLP, PEFT.

Cloud
AWS (S3, SageMaker, EC2, Lambda, ECR, EKS, etc.),

Azure (Blob Storage, ML, OpenAI, Language, ACR, AKS).

MLOps
MLflow, DVC, Neptune.

Backend
FastAPI.

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Data Engineering
Apache Airflow.

Databases
PostgreSQL, MongoDB, ChromaDB, Pinecone.

DevOps
Docker, Docker Compose, Kubernetes(k8s).

Software Development
Pynput.

Source control systems
Git, Github.

Projects

CHATBOT FOR SALES STAFF

The goal of this project is to optimize the performance of the sales team in an IT company by analyzing the dialog with the client and assisting with selecting the most relevant products/candidates.

Project roles
Data Scientist / Machine Learning Engineer

Period
09.2023 – till now

Responsibilities & achievements
- Collected and prepared data for training different models;
- Experimented with different LLMs for various different tasks (NER, summarization, key phrase extraction, etc.);
- Set up a vector database using ChromaDB for storing client requests and employee information ;
- Experimented with various embedding models;
- Created a data extraction and preprocessing pipeline;
- Implemented data processing layer for employees to extract useful insights and vectorize the person for the recommendation engine;

- Created custom Langchain agents to complete different tasks based on the bot functionality(q&a, recommendation query formulation, entity extraction, etc.);
- Implemented a recommendation system that extracts the most appropriate candidates for the job description;
- Added speech-to-text model to the preprocessing pipeline;
- Deployed models to Azure.

Environment

Python, Numpy, Matplotlib, Pandas, Seaborn, PyTorch, Hugging Face, BitsAndBytes, OpenAI API, Langchain, Llama-index, PEFT, FastAPI, MongoDB, Pinecone, ChromaDB, Docker, Docker Compose, Github, Azure (Blob Storage, ML, OpenAI, Language, ACR, AKS).

PROMPT-BASED AUTOMATION ENGINE

POC for RPA engine that records and executes prompt-based tasks. The goal of this project was to create a system that allows users to record certain tasks based on a prompt and then the system can execute the same task but with different context.

Project roles

Data scientist / Machine Learning engineer

Period

10.2022 – 09.2023

Responsibilities & achievements

- Implemented transcription and execution pipeline using Pynput;
- Trained BERT models in Azure Language Studio for classifying prompts and extracting useful entities and phrases;
- Setted up ML experiments tracking using MLflow;
- Experimented with different detection models and approaches to find useful ui elements from screenshots;
- Tried different sentence comparison approaches to identify which is the best for task matching;

- Experimented with different NLP approaches for entity matching and specification fill-out;
- Integrated the NLP pipelines into the RPA engine;
- Setup schema completion using GPT-3.5;
- Setup monitoring tools to track various metrics like model metrics, response time and resource utilization;
- Created LLM agents using Langchain for prompt classification, schema completion and question answering;
- Integrated all pipelines into the implemented recording and executing engine;

Environment

Python, PyTorch, Hugging Face, OpenCV, Ultralytics, Tesseract OCR, NLTK, OpenAI API, Spacy, Langchain, CoreNLP, MLflow, DVC, MongoDB, ChromaDB, Docker, Docker Compose, Kubernetes(k8s), Pynput, GitHub, Git, Azure (Blob Storage, Machine Learning, Language, VM, Container Apps, AKS, OpenAI).

SALES PREDICTION ENGINE

An analytics engine for sales volume prediction is a tool that uses historical sales data and other relevant information to predict future sales volumes for a particular product or group of products. The engine typically employs advanced statistical techniques and machine learning algorithms to analyze the data and make accurate predictions.

Project roles

Data scientist / Machine Learning engineer

Period

04.2021 – 10.2022

Responsibilities & achievements

- Conducted in depth exploratory data analysis with an eye to derive initial hypothesis and core issues to tackle later on;
- Developed temporal features, including lag variables, to capture time-based patterns in transactional data;
- Integrated an analytics layer using Apache Spark within the engine architecture, emphasizing

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scalability to proficiently manage surges in data volumes and dynamic shifts in business needs;
- Created a statistical outlier detection layer that improved overall ML model performance and delivered various insights;
- Proposed and integrated a recommendation system that made relevant advice for the end businesses and optimized their buys;
- Developed and implemented an anomaly detection system, utilizing machine learning techniques to identify and flag irregularities in sales data;
- Trained and tested multiple TS approaches like ARIMA family, Facebook Prophet, GBDT algorithms;
- Implemented interpretability layers to enhance transparency and explainability of model predictions;
- Implemented Boruta within the feature engineering pipeline to enhance model interpretability and streamline the predictive modeling process.
- Deployed models on AWS SageMaker;
- Implemented Apache Airflow for orchestrating end-to-end workflows in the sales volume prediction project;
- Delivered useful analytical reports to the end customer.

Environment

Python, Numpy, Pandas, Seaborn, Scikit-learn, XGBoost, LightGBM, Prophet, Boruta, Neptune, Apache Airflow, PostgreSQL, Docker, Docker Compose, Github, AWS (S3, SageMaker, EC2, Lambda, ECR, EKS, etc.).

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	Python	3	2024
Data Science	Numpy	3	2024
	Pandas	3	2024
	Matplotlib	1	2024
	Seaborn	3	2024
Machine Learning	Scikit-learn	2	2022
	XGBoost	2	2022
	LightGBM	2	2022
	Prophet	2	2022
	Boruta	2	2022
Deep Learning	PyTorch	2	2024
	Hugging Face	2	2024
	BitsAndBytes	1	2024
Computer Vision	OpenCV	1	2023
	Tesseract OCR	1	2023
	Ultralytics	1	2023
Natural Language Processing	NLTK	1	2023
	OpenAI API	2	2024
	Spacy	1	2023
	Langchain	2	2024
	LlamaIndex	1	2024
	CoreNLP	1	2023

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	PEFT	1	2024
Cloud	AWS	3	2022
	Azure	3	2024
MLOps	MLflow	1	2023
	DVC	1	2023
	Neptune	2	2022
Backend	FastAPI	1	2024
Data Engineering	Apache Airflow	2	2022
Databases	PostgreSQL	2	2022
	MongoDB	2	2024
	ChromaDB	2	2024
	Pinecone	1	2024
DevOps	Docker	3	2024
	Docker Compose	3	2024
	Kubernetes(k8s)	1	2023
Software Development	Pynput	1	2023
Source control systems	Git	1	2023
	Github	3	2024